

Yiyao Hu

Machine Learning Engineer | Recommendation Systems & LLM Applications

📞 (626)267-8912 @ ionahu08@gmail.com 🔗 <https://www.linkedin.com/in/yiyao-hu/> 🔗 <https://github.com/ionahu08> 📍 Sunnyvale, CA

SUMMARY

Machine Learning Engineer with a master's degree in Biostatistics (UCLA) and 4 years in FDA-regulated pharmaceutical research, where I built validated data pipelines for large-scale clinical studies. I now bring that statistical and data-engineering rigor to end-to-end ML systems — spanning LLM- and agent-powered applications and multi-stage recommendation systems — cutting cost and latency while staying reliable in production. I'm looking for engineering roles where I can own systems end to end, from data and modeling to production.

PROJECTS

[LLM&Agents] NomNom—Food Analysis & Health Tracking App with Multimodal AI Agents

📅 03/2026 - Present

🔗 <https://github.com/ionahu08/nomnom-ai>

- Built NomNom, an AI-powered food tracking app that analyzes meal photos using **multimodal AI** to estimate nutrition and generate personalized diet recommendations.
- Architected a full-stack system (FastAPI + SwiftUI) with **task-based model routing** (Haiku for lightweight extraction, Sonnet for complex reasoning) and **orchestration** (tool use, retry, fallback), achieving **5-15s latency** and **4.3x cost reduction** (\$12 → \$2.80/user/day) vs. a Sonnet-only baseline.
- Designed a **multi-agent architecture** combining a fixed orchestrator-worker pipeline (vision → nutrition → recommendation) for structured analysis with an autonomous **plan-act-reflect loop** for the interactive coach, choosing each pattern by task to balance reliability and flexibility.
- Engineered a **production-grade LLM harness** with **prompt templating**, structured outputs, and validation + retry **guardrails**, achieving a **near-100% success rate** on 10K+ test API calls, driving parsing failures to near-zero.
- Built a semantic caching system with **RAG-style retrieval** using pgvector **embeddings** and **hybrid search (vector + BM25)**, achieving an **85% cache hit rate** through an empirically-tuned similarity threshold (0.82, tested on 150 real meals), sharply cutting redundant LLM calls.

[RecSys] Multi-Stage Live Video Recommendation System (KuaiLive)

📅 01/2026 - 03/2026

🔗 https://github.com/ionahu08/kuaishou_live_room_multi_stage_ranking

- Built a **multi-stage recommendation system** (retrieval → ranking → reranking) to optimize **user engagement** (click, watch-time, like) for a live video platform, using 5.36M interactions, 23K users, 450K streamers, and 11.6M+ rooms.
- Designed **large-scale feature pipelines** (user, item, context, behavior) with strict time-based splits, deduplication, and leakage control (12M+ samples), enabling realistic offline evaluation; optimized training and evaluation with recency weighting to better reflect real-world user behavior.
- Integrated **CLIP-based visual encoders** to extract **image embeddings** from video covers, enhancing **multimodal feature representation** for both retrieval and ranking models.
- Trained a **dual-encoder (two-tower) retrieval model** with **contrastive + BCE-with-logits loss** and **recency-weighted sampling**; deployed **FAISS ANN** for top-K retrieval (Recall@100: 0.391).
- Trained a multi-task **DCNv2 ranker** and built an end-to-end evaluation framework with **MMR-based reranking** to improve relevance and diversity, achieving **AUC 0.946 / 0.923**, **Precision@10 0.439**, and stable NDCG@k under strict time-based, leakage-controlled splits.

[RecSys] DCNv2-based Cold-Start CTR Ranking for Micro-Video Ads

📅 12/2025 - 01/2026

🔗 https://github.com/ionahu08/DCNv2_with_PyTorch

- Built an end-to-end **CTR ranking system using DCNv2** to predict ad click-through probability, combining **low-rank CrossNetV2** (explicit feature interactions) and a **deep MLP** (nonlinear representation learning).
- Trained on 12.7M interactions (10.9K users, 1.7M videos) using a **strict cold-start item split (no item overlap)** to evaluate generalization to unseen ads, reflecting real-world cold-start ad ranking constraints.
- Built multimodal features fusing sequential user behavior and temporal context with 64-d image embeddings (clipping / log-scaling for stability).
- Achieved strong cold-start performance: **AUC ≈ 0.71**, **Precision@100 ≈ 0.27**, **Recall@100 ≈ 0.64**, **NDCG@100 ≈ 0.51**.

EDUCATION

Master's Degree, Biostatistics

University of California, Los Angeles

📅 09/2020 - 06/2022

- GPA 3.89/4.0

Bachelor of Medical Sciences

Chongqing Medical University, China

📅 09/2014 - 07/2019

- GPA: 88.13/100 (Rank: 1/49)

EXPERIENCE

Senior Statistical Programmer

Arrowhead Pharmaceuticals

📅 06/2022 - 02/2026 📍 Pasadena, CA

- Delivered **10+ clinical studies across Phase I-III**, owning end-to-end clinical data transformation, validation, and reporting — including a lead compound taken through **FDA submission**.
- Built a parameterized, **scalable SAS / SQL templating framework** for standardized report generation, reusable across studies and adopted by the full **8-person team** — cutting output production from **~2 hours to ~20 minutes** and setting the team's first consistent output standard.
- Established rigorous **data validation and quality checks** (independent double programming, automated cross-checks) ensuring submission-grade data quality for downstream analysis and reporting.
- Designed a **cross-study tracking system** spanning **4 departments**, shifting the team from reactive firefighting to proactive planning with data-backed timeline negotiations.
- Led planning for a report deliverable under a tight **3-week deadline (9 studies, 3 people)**, with a risk-tiered phased plan that won **skip-level approval** and cut **per-person workload ~40%**; mentored 2 new hires.

Statistical Programmer Intern

Gilead Sciences

📅 06/2021 - 08/2021 📍 Foster City, CA

- Performed ETL and data validation on 20+ clinical trial datasets, improving data quality and reproducibility for downstream analysis and regulatory reporting.

SKILLS

Machine Learning

Multi-Stage Retrieval & Ranking, Two-Tower Models, DCNv2, DeepFM, CTR Prediction, Multimodal Learning, A/B Testing

LLM & AI Systems

LLM Harness Engineering, Agentic Workflows, RAG, Prompt Engineering & Evaluation, Transformer, MCP, Claude API, Claude Code

Programming & Frameworks

Python, SQL, SAS, R, PyTorch, scikit-learn, Pandas, NumPy, FastAPI · Data Structures & Algorithms

Systems & Infrastructure

Vector Search (pgvector, FAISS), GPU Training, Cloud / GPU (Lambda Labs, Vast.ai), API Design

Certification

Certified Claude Architect (Anthropic)